

Common N-Grams as Causal Probes of Transformer Computation: Attention Signatures, MLP Compilation, and Residual Transformation Trajectories

Paper 0.5 — Continuous Self-Attention Learning Research Line

Controlled experiment report — August 2026

Abstract

Transformers repeatedly encounter short token sequences whose continuation distributions can be estimated directly. We use recurrent prefix relations to test component responsibility, predictive equivalence, and nuisance-conditioned variance across self-attention (SA), feed-forward/MLP layers, and the residual stream. Two locally trained causal Transformers (two seeds each) yield 1,920 component rows and a controlled depth–prefix–noise factorial. Under pattern evidence with four distractors, post-block JS dispersion falls by 0.0474 bits from first to last available depth (family/run bootstrap 95% CI $[-0.0778, -0.0217]$), even while within-example entropy rises. Stable attention motifs remain unrelated to SA causal contribution ($\rho = -0.009$). We then replicate on Pythia-70M-deduped checkpoints 0, 1,000, and 143,000 and pinned Qwen3-0.6B using crossed syntax–semantic families. Directions fitted on clean weak-evidence identities transfer to noisy stronger-evidence identities. Equivalent replacement is substantially less harmful than fully nonequivalent replacement for trained Pythia at step 1,000 (difference 1.047 nats, family bootstrap CI $[0.545, 1.455]$), step 143,000 (1.206, $[0.224, 2.187]$), and Qwen (0.599, $[0.299, 0.899]$), but not at random initialization. Removing the fitted common direction is harmful in the trained models. Prediction-dispersion reduction is not universal across pretrained depth: final Pythia shows a small negative contrast, whereas Qwen increases. The supported claim is therefore causal functional equivalence with conditional nuisance refinement, not monotone denoising or online learnability. E7 remains stopped.

1 Motivation

A language model trained by next-token prediction is exposed to a vast collection of recurrent local relations. For a token prefix

$$g = (x_{t-n+1}, \dots, x_t), \quad (1)$$

the training distribution defines a continuation distribution

$$P_{\mathcal{D}}(x_{t+1} \mid g). \quad (2)$$

Common n-grams are useful mechanistic probes because both the input relation and the statistical target are measurable. Unlike broad semantic concepts, they provide a compact unit for asking where a known conditional relation is represented and how it is transformed.

The central question is not merely whether the model predicts the continuation correctly. It is how responsibility is divided among three interacting mechanisms:

1. **Self-attention:** context-dependent selection, matching, retrieval, and relational binding;
2. **MLP/feed-forward layers:** nonlinear transformation and possible storage of learned associations;
3. **Residual stream:** accumulation, refinement, competition, repair, and transport of partial representations across depth.

This decomposition connects two broader research lines. First, it gives the residual-stream program a controlled family of transformations for measuring refinement, complementarity, interference, and repair. Second, it provides a minimal laboratory for continuous learning from SA statistics: recurrent context computations may expose stable relations that can later be compiled into persistent neural structure.

2 Relation to Prior Work

Several literatures motivate the proposed study, but they do not fully characterize the joint SA–MLP–residual trajectory of corpus n-grams.

MLPs as associative memories. Geva et al. showed that transformer feed-forward layers can be interpreted as key–value memories: keys correlate with recognizable textual patterns and values induce output-vocabulary distributions. Lower layers tend to capture shallower patterns while upper layers capture more semantic patterns, and FFN outputs are subsequently refined through residual computation [1].

Attention circuits and induction. Mechanistic work on attention has identified copying, previous-token, prefix-matching, and induction-head circuits that dynamically recover relations from the current context [2]. These mechanisms show that a prefix-to-continuation relation need not be stored as a fixed lexical lookup in MLP weights; it can be constructed or retrieved from context.

Specialization of attention and MLP computation. Controlled studies comparing trainable and frozen transformer components indicate that MLPs are especially effective for memorization, while input-dependent attention is important for retrieval and induction-like behavior [3]. This suggests a division of labor, but real language modeling is likely to contain mixed and distributed circuits rather than a strict dichotomy.

Bigram subnetworks. Chang and Bergen identify small subnetworks sufficient for basic bigram predictions, concentrated in the first transformer MLP layer. These subnetworks can drive an early residual transition from representing the current token toward predicting the next token [4]. This result makes ordinary n-gram statistics especially attractive as a basis for extending analysis beyond bigrams to context-dependent and compositional sequences.

Gap addressed here. We propose a unified experimental framework in which corpus statistics, attention structure, component-wise residual updates, causal intervention, and training-time emergence are measured together. The target is not a slogan such as “attention retrieves, MLP memorizes,” but a quantitative map of when each mechanism dominates, cooperates, conflicts, or repairs another.

3 Core Hypotheses

Let g be an n -gram prefix and y a likely continuation. We distinguish five hypotheses.

3.1 H1: Direct MLP compilation

For highly frequent, low-entropy lexical relations, earlier residual computation produces a representation z_g that an MLP maps strongly toward the continuation:

$$z_g \xrightarrow{\text{MLP}_\ell} v_y. \quad (3)$$

Prediction gains should be localized in one or a few MLP updates and remain relatively stable across surrounding contexts.

3.2 H2: Attention-dominant reconstruction

For relations that depend on information introduced or disambiguated in the current context, attention should provide a larger causal contribution by selecting and combining relevant prefix elements or prior occurrences. Induction-style synthetic controls are the limiting case.

3.3 H3: Distributed SA–MLP circuits

Many relations will be distributed. An attention layer may retrieve a relevant feature, an MLP may transform it toward a continuation, and later layers may combine, sharpen, suppress, or contextualize it. The mechanistic object of interest is therefore a trajectory rather than a single neuron or head.

3.4 H4: Statistical compilation with training

Repeated relations may migrate during training from diffuse contextual computation toward more direct and economical mappings. If so, training partially acts as amortized compilation of recurrent inference.

3.5 H5: Contextual override and repair

Strong lexical priors can be contradicted by wider context. For example, a common continuation may be promoted early and subsequently suppressed. Such cases provide controlled probes of residual interference, inhibition, and repair.

4 N-Gram Atlas

We construct an atlas from one or more public corpora tokenized with the target model tokenizer. For each prefix g record:

$$f(g) = \text{corpus frequency}, \quad (4)$$

$$H(g) = - \sum_y P(y | g) \log P(y | g), \quad (5)$$

$$p_{\max}(g) = \max_y P(y | g), \quad (6)$$

$$\text{PMI}(g, y) = \log \frac{P(y | g)}{P(y)}. \quad (7)$$

The main strata are:

- high-frequency / low-entropy lexical sequences;
- high-frequency / high-entropy prefixes;
- rare but deterministic sequences;
- same suffix with different earlier prefixes and different continuations;
- same relation under lexical substitutions;
- compositional/synthetic sequences absent or rare in pretraining-like corpora;
- context-introduced mappings for induction controls;
- override examples where wider context contradicts the dominant local n-gram continuation.

N-gram lengths $n \in \{1, 2, 3, 4, 5, 8\}$ provide a progression from local token association to multi-token contextual binding. Tokenizer effects must be retained rather than hidden: the study concerns token n-grams, with optional word-level groupings as a secondary analysis.

5 Layer-wise Instrumentation

For a pre-norm decoder block, record the residual state before attention r_ℓ , the attention update Δ_ℓ^{SA} , the post-attention residual r'_ℓ , and the MLP update Δ_ℓ^{FF} :

$$r'_\ell = r_\ell + \Delta_\ell^{SA}, \quad (8)$$

$$r_{\ell+1} = r'_\ell + \Delta_\ell^{FF}. \quad (9)$$

Where architecture permits, also record per-head updates, MLP intermediate activations, normalized states, Q/K/V projections, and post-RoPE Q/K values.

5.1 Logit-space progress

Using the model’s unembedding matrix W_U , define a diagnostic projection

$$\ell_y(r) = [W_U \text{Norm}(r)]_y. \quad (10)$$

For the target continuation y , measure component-wise gains

$$U_\ell^{SA}(y) = \ell_y(r'_\ell) - \ell_y(r_\ell), \quad (11)$$

$$U_\ell^{FF}(y) = \ell_y(r_{\ell+1}) - \ell_y(r'_\ell). \quad (12)$$

These quantities are descriptive and must be complemented by causal interventions because logit-lens projections can misrepresent intermediate computation.

5.2 Causal contribution

For each component, compare the intact model with interventions such as zero ablation, mean ablation, path patching, activation replacement from matched controls, and selective head/MLP ablations. Let

$$C_{c,\ell}(g, y) = \log P(y \mid g)_{\text{intact}} - \log P(y \mid g)_{\text{intervene}(c,\ell)}. \quad (13)$$

Matched semantic/context controls prevent attributing generic language-model utility to the n-gram relation itself.

6 Attention Signatures as Relational Invariants

For head h at layer ℓ , let $A^{\ell,h}$ denote the attention matrix. For occurrences of g , extract the submatrix spanning the n-gram tokens and relevant context. Define the conditional motif

$$\bar{A}_g^{\ell,h} = \mathbb{E}[A^{\ell,h} \mid g] \quad (14)$$

and a stability statistic such as

$$S_A(g, \ell, h) = 1 - \mathbb{E} \left[d(A_{g,i}^{\ell,h}, \bar{A}_g^{\ell,h}) \right], \quad (15)$$

where d can be Frobenius distance after normalization, cosine distance, Jensen–Shannon divergence over rows, or a graph-topological distance.

The key question is whether exact activation coordinates vary while a relational pattern remains stable. We therefore compare:

- raw attention matrices;
- row-normalized motif geometry;
- attention graph edges above adaptive thresholds;
- connected components / communities;
- QK similarity structure;
- OV contribution directions;
- residual-update direction and subspace similarity.

This turns “representation” from a single-vector question into a relational invariance question.

7 Mechanistic Trajectories

For each (g, y) define a depth trajectory

$$\mathbf{T}(g, y) = [U_1^{SA}, U_1^{FF}, \dots, U_L^{SA}, U_L^{FF}, C_{SA,1}, C_{FF,1}, \dots]. \quad (16)$$

Cluster trajectories and test whether clusters correspond to measurable corpus properties. Candidate classes include:

Early lookup:

a sharp early MLP contribution dominates;

Progressive composition:

alternating SA and MLP contributions accumulate gradually;

Late disambiguation:

several candidates coexist until a later context-sensitive update;

Retrieval/copy:

attention dominates and MLP contribution is weak;

Override/repair:

an early common continuation is promoted, then later suppressed in favor of a contextually correct one.

A central quantitative analysis predicts trajectory class, layer of maximal causal contribution, and SA/MLP contribution ratio from $\log f(g)$, entropy $H(g)$, PMI, n-gram length, and context sensitivity.

8 Training-Time Emergence

Tiny and small language models should be trained from scratch with checkpoints saved densely enough to follow mechanism formation. At training step t , measure

$$\mathbf{T}_t(g, y), \quad S_{A,t}(g, \ell, h), \quad C_{c,\ell,t}(g, y). \quad (17)$$

We test whether frequent deterministic relations show a sequence such as

$$\text{diffuse contextual computation} \rightarrow \text{stable relational motif} \rightarrow \text{localized/compiled mapping}. \quad (18)$$

A negative result is equally informative: mechanisms may remain distributed, or early MLP bigram mappings may appear before stable attention motifs. The study must not assume compilation direction in advance.

9 Bridge to Continuous Learning from SA Statistics

The later research line asks whether useful regularities visible during inference can be retained without full conventional retraining. Paper 0.5 does not claim to solve this problem. It defines the measurable precursor.

Suppose a recurring context relation produces samples of attention/residual structure $s_1, \dots, s_m$ and continuations $y_1, \dots, y_m$. An online system can maintain sufficient statistics or prototypes

$$\theta_g^{\text{stat}} = \{N_g, \bar{s}_g, \Sigma_g, \hat{P}(y | g), \bar{\Delta}r_g\}. \quad (19)$$

If the relation is stable and predictive, a later algorithm may convert it into persistent structure:

$$\text{recurrent SA relation} \rightarrow \text{stable prototype/statistic} \rightarrow \text{adapter or weight memory}. \quad (20)$$

Paper 0.5 therefore establishes candidate signals, stability criteria, causal targets, and baselines for a later “neural compilation” experiment. The online learner should be evaluated by whether it reproduces or improves the native trajectory while reducing repeated contextual work, not merely by whether loss decreases after adaptation.

10 Experiments

10.1 E1: Corpus atlas and behavioral predictability

The committed atlas builder counts continuations by brute-force-equivalent sliding windows for $n \in \{1, 2, 3, 4, 5, 8\}$. Each row records token IDs, decoded inspection text, exact frequency, continuation counts/probabilities, entropy, top probability, log association ratio, source/split, and a content hash. Deterministic strata separate frequency and entropy. In this local controlled experiment the corpus is the true training corpus; the implementation nevertheless uses the field name `reference_corpus_frequency` so the schema transfers safely to opaque pretrained models.

10.2 E2: SA versus MLP contribution across depth

An inspectable pre-normalized causal Transformer records r_ℓ , Δ_ℓ^{SA} , r'_ℓ , Δ_ℓ^{FF} , $r_{\ell+1}$, and per-head attention. Capture mode preserves bit-identical logits because attention probabilities are reconstructed separately from the output kernel. Signed diagnostic progress is reported separately from causal log-probability drops. Interventions include zero update, batch-matched mean update, matched-control activation replacement, first-head ablation, and full FFN-layer ablation. Residual identities and trace-disabled parity are unit tested.

10.3 E3: Stable attention motifs

For every probe and layer, the final-query attention rows are head-aligned and row normalized. Within-relation cosine is compared with a position-matched control that changes the defining prefix or contextual mapping. The decisive follow-up correlates relation-layer motif specificity with SA zero-ablation utility; visual similarity alone is not treated as evidence.

10.4 E4: Stored versus context-introduced relations

The controlled factorial comparison includes a high-frequency deterministic mapping $(1, 2) \mapsto 3$, a high-frequency high-entropy mapping $(4, 5) \mapsto \{6, 7, 8\}$, a rare deterministic mapping, and arbitrary in-context definitions of the form $(A, B), \dots, A \mapsto B$. The latter mappings vary their arbitrary labels while preserving the relational template.

10.5 E5: Override and repair

A leading marker changes the otherwise habitual continuation $(1, 2) \mapsto 3$ to the correct continuation 13. At every sublayer we retain signed progress for both candidates, causal drops, update–update cosine, and a geometry-only candidate label. The labels *opposing*, *aligned*, and *orthogonal candidate* deliberately do not assert interference without negative task utility and later repair.

10.6 E6: Training-time development

Two model settings (width/layers/heads = 32/2/2 and 48/3/3) are trained for 160 optimizer steps with seeds 11 and 23. Component measurements are repeated at steps 0, 40, 100, and 160; the complete four-checkpoint state series is committed for the first setting/seed. This tests developmental direction without assuming monotone migration toward the FFN.

10.7 E7: Minimal online compilation pilot

E7 was not started. The near-zero motif–causal association and the absence of pretrained replication fail the pre-registered gate for choosing SA statistics as an online-learning signal. Preserving this stop decision prevents a learner from being engineered around a preferred interpretation of an inconclusive pilot.

11 Models and Data

The present experiment uses the two locally trained decoder-only settings above, a 40-token vocabulary, maximum sequence length 12, GELU FFNs, AdamW, and 960 controlled training sequences per seed. Probe contexts cover four primary strata with 24 examples each. The local regime makes training frequency and checkpoints exact and keeps exhaustive intervention feasible. It is intentionally not a substitute for natural-corpus or pretrained-model replication; GPT-2/Pythia-style checkpoint series and a modern small decoder remain the next external-validity stage.

12 Metrics and Statistical Tests

Primary metrics include:

- target log-probability and rank across depth;
- SA and MLP logit contribution curves;
- causal effect under component interventions;
- attention-motif stability across occurrences;
- residual-update cosine/subspace similarity;
- layer of maximal contribution;
- contribution concentration (e.g., entropy across layers/components);
- trajectory-cluster stability across seeds and models;
- regression of mechanism metrics on frequency, continuation entropy, PMI, and n-gram length.

Report bootstrap confidence intervals over n-grams and contexts, not just token occurrences. Mixed-effects models can separate effects of n-gram identity, model seed, and context. Multiple-comparison control is required for head-level claims.

12.1 Conditional realization-variance experiment

Variance reduction is defined across independently constructed token-sequence realizations conditional on the same relation, never across layers. Depth is the refinement axis. For every probe relation we cross weak (one-token), pattern-length (two-token), and full-context evidence with zero, two, or four deterministic nuisance distractors. At every pre-SA, post-SA, and post-MLP location we estimate target-probability, target-logit, and target-margin variance; mean JS divergence to the relation-conditioned output centroid; residual covariance trace/effective rank; within-example entropy; and between-relation/within-relation JS separation. Contradictory override contexts remain a separate condition. Success does not require monotonicity: the primary prediction is rising or stable signal with lower nuisance-conditioned dispersion over useful depth.

13 Controlled Pilot Results

The run produced 21,630 atlas rows, 1,920 final-checkpoint component rows, 1,920 raw motif/control rows, 2,880 entropy-trajectory rows, 3,840 causal equivalence-patch rows, and four dense checkpoints for the first model/seed. Final training losses range from 2.436 to 2.565. All numbers below regenerate from the committed manifest and tables.

Stratum	Component	Zero	Mean	Replacement	Head 0
Context introduced	FF	0.827	0.292	0.784	–
Context introduced	SA	0.305	0.120	0.565	0.855
Familiar, high entropy	FF	0.942	0.421	0.642	–
Familiar, high entropy	SA	1.027	0.069	0.998	0.158
Familiar, low entropy	FF	0.478	0.442	0.134	–
Familiar, low entropy	SA	0.301	0.025	0.407	0.319
Override/repair	FF	0.261	0.450	0.205	–
Override/repair	SA	1.497	0.347	3.323	0.639

Table 1: Mean intact-minus-intervened target log probability (nats). Positive values indicate target-supporting causal contribution. Rows aggregate layers, model settings, seeds, and n-gram examples; scientific-unit uncertainty is reported in the machine-readable component table. Head 0 applies only to SA.

The intervention family gives a more qualified result than either direct compilation or attention-dominant retrieval. Familiar low-entropy mappings are FFN-heavier under zero ablation (0.478 versus 0.301 nats), but the effect changes under matched replacement (0.134 versus 0.407). Context-introduced mappings are also FFN-heavier under zero ablation (0.827 versus 0.305), contrary to H2’s simplest prediction, while their first-head intervention is substantial (0.855). High-entropy familiar mappings use both components (SA 1.027; FFN 0.942). Override/repair is the clearest SA-dominant stratum (1.497 versus 0.261) and has the largest matched-replacement SA effect (3.323). These method-dependent differences support distributed circuits and make generic component importance an active alternative explanation.

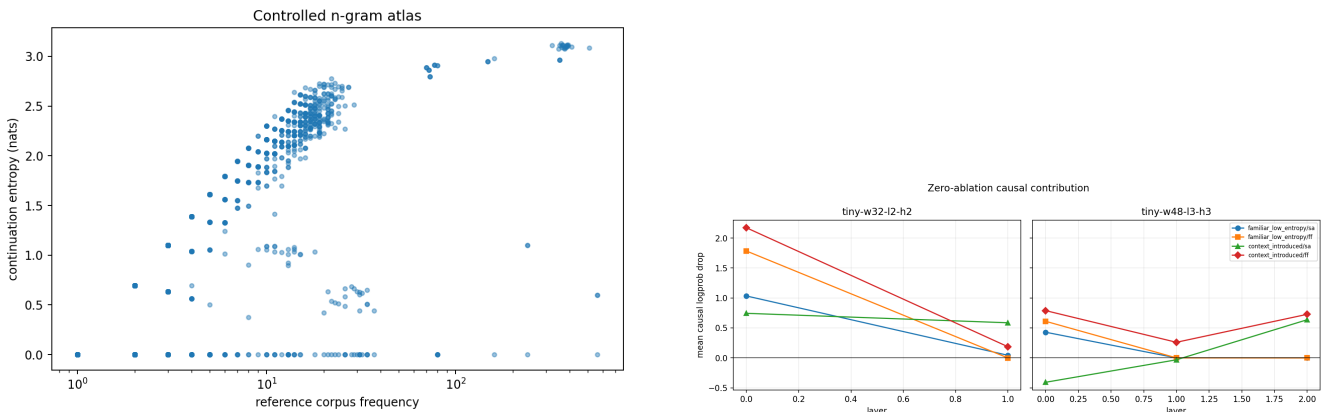


Figure 1: Left: controlled atlas frequency–entropy support. Right: layerwise zero-ablation effects for the two model settings. Curves include all pre-registered strata; the displayed comparison emphasizes familiar low-entropy and context-introduced mappings.

13.1 Externally defined predictive equivalence survives causal replacement

We define equivalence from the controlled continuation relation, before inspecting hidden states. Model output distributions for two contexts in the same relation family have mean Jensen–Shannon (JS) divergence 0.028 bits and top-5 overlap 0.878, compared with 0.494 bits and 0.645 for each context’s prefix-matched nonequivalent control. This separation is therefore functional rather than a cluster chosen from the representation under test.

Replacement interventions sharpen the result. Across layers and components, equivalent-donor replacements produce far smaller output changes than nonequivalent donors. The contrast is largest at layer-0 SA (0.028 versus 0.436 JS bits); its mean target-probability change is +0.003 for equivalent donors and −0.352 for nonequivalent donors.

Layer-0 FFN shows the same ordering (0.017 versus 0.156 JS bits). Later-layer contrasts are smaller but retain the ordering. Because complete update tensors are replaced, this supports transferable family-conditioned computation in this controlled regime, while not identifying a unique direction or proving natural-language generality.

The entropy trajectories are explicitly non-monotone and task-conditioned. Mean decoded entropy at pre-SA/post-SA/post-block is 0.069/0.046/0.083 bits for familiar deterministic relations, but 0.637/0.809/1.198 for high-entropy relations and 0.848/1.158/1.959 for context-introduced mappings. Thus SA and FFN can increase uncertainty when the learned continuation family requires alternatives; entropy reduction is not treated as a universal signature of useful computation. Family-conditioned update centroids are highly aligned in many cells, but their effective ranks and within-family dispersion remain in the artifact, so commonality is not collapsed into a one-dimensional mechanism claim.

13.2 Depth reduces nuisance-conditioned prediction dispersion

The completed variance grid contains 124,416 fixed-depth realization rows and 7,776 conditional aggregates across trained/random stages, context controls, and observed/shuffled targets. Under pattern-length evidence with four distractors, mean post-block target-probability variance falls from 0.0347 at layer 0 to 0.0106 at layer 1 and 0.0109 at layer 2. Mean JS-to-pattern-centroid dispersion similarly falls from 0.0650 to 0.0307 and 0.0106 bits. The median between-pattern/within-pattern JS ratio rises from 56.9 to 142.0 to 285.2 across the available depth cells. This is evidence for nuisance invariance as repeated state-conditioned refinement, not evidence that layers are independent samples.

The distinction from entropy matters empirically. Over the same cells, mean within-example entropy rises from 1.326 to 2.094 to 2.277 bits while across-realization JS dispersion falls. Depth therefore makes outputs more consistent across irrelevant realizations without forcing a narrower distribution within each realization. Added nuisance increases post-block JS dispersion for pattern evidence (0.0119/0.0263/0.0404 bits at zero/two/four distractors, averaged across depth and runs), and later computation partly compensates. Weak-prefix cells include exact-output ties and near-zero within-family denominators, so SNR means are unstable; we report medians and retain the raw numerator and denominator.

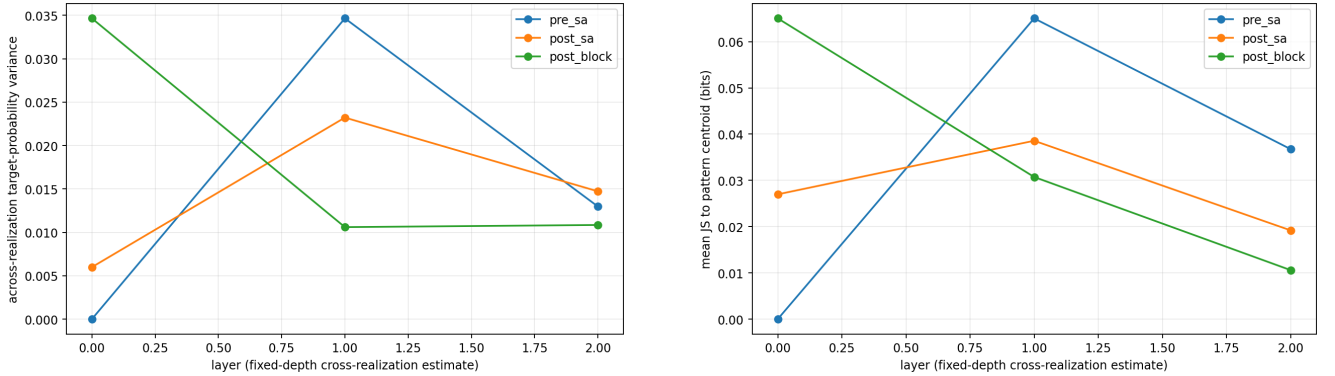


Figure 2: Across-realization dispersion at fixed depth under pattern-length evidence and four distractors. Left: target-probability variance. Right: full-distribution JS dispersion. These are conditional sample statistics, not variance across layers.

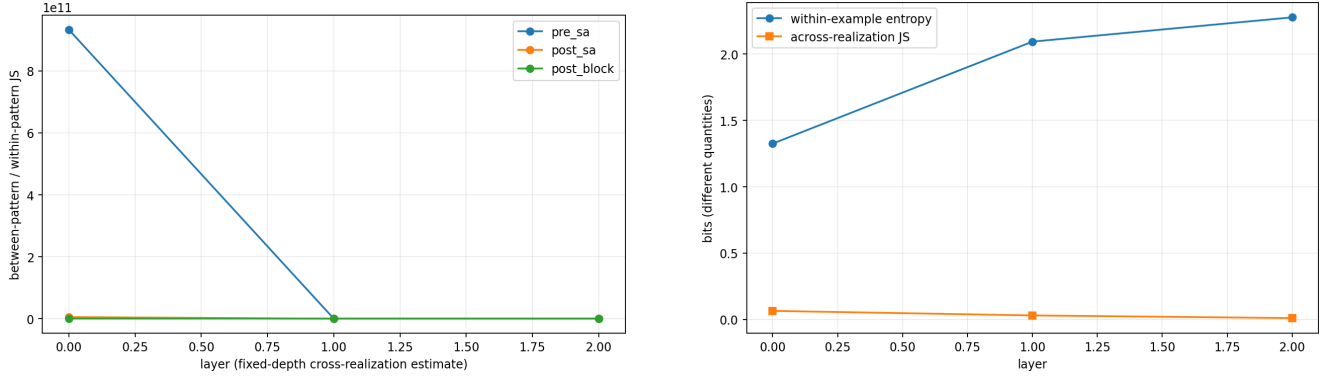


Figure 3: Left: between-pattern/within-pattern separation (curves plot cell means; robust medians are used in prose). Right: within-example entropy and across-realization JS are distinct and can move oppositely.

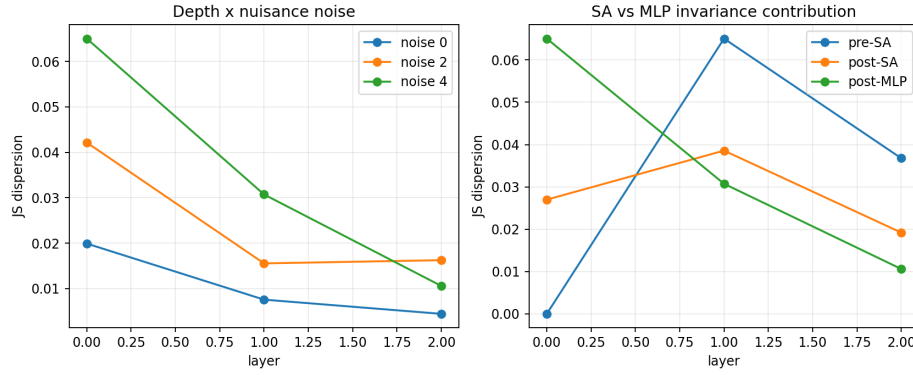


Figure 4: Depth-by-noise dispersion and pre-SA/post-SA/post-MLP comparison. Sublayer changes are interpreted as contributions to invariance, not independent estimators.

The completed factorial retains all prefix strengths and distinguishes irrelevant nuisance, answer-changing contexts, shuffled target labels, and a true step-0 checkpoint. For pattern evidence with four distractors, trained irrelevant- nuisance JS falls by -0.0474 bits from first to last depth (24 relation-run units; family/run bootstrap CI $[-0.0778, -0.0217]$). Answer-changing contexts also contract (-0.0328 , $[-0.0543, -0.0141]$), but less strongly, so invariance is not equated with ignoring legitimate context. The random checkpoint has extremely low dispersion only because target signal is absent (mean target probability below 5×10^{-7}); low variance without signal is not success. Shuffling the target changes target-specific variance/margin summaries but correctly leaves full-distribution JS unchanged.

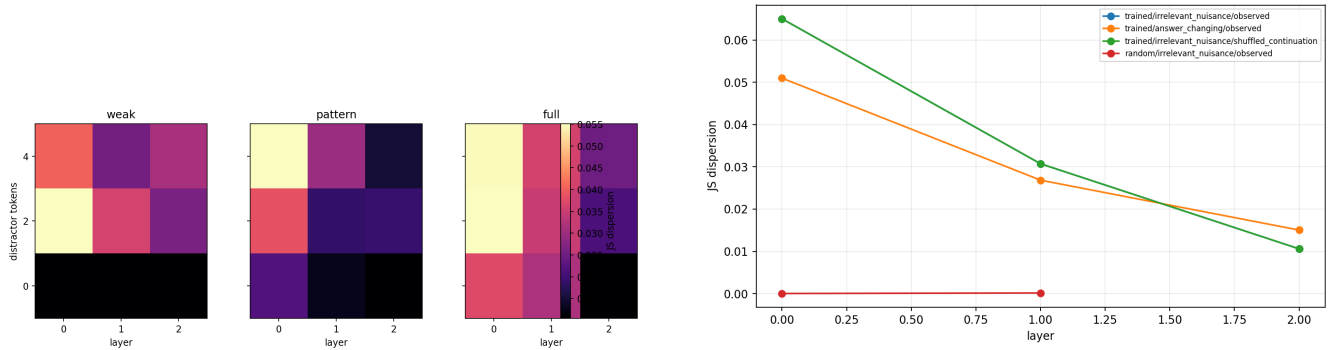


Figure 5: Left: complete depth-prefix-noise JS surface. Right: irrelevant, answer-changing, shuffled-target, and random-checkpoint controls. Low random-model dispersion is interpreted jointly with its absent target signal.

13.3 Pretrained checkpoint and modern-model replication

The external-validity matrix uses Pythia-70M-deduped at official revisions `step0`, `step1000`, and `step143000` [5], plus Qwen3-0.6B at pinned revision `c1899de...` [6]. Four externally defined families cross alternation/mapping syntax with number/color semantic domains. Each family crosses one, two, or four demonstrations; zero, two, or four distractor units; and four realizations. Selected first/middle/final depths yield 1,728 observation rows. Exact full-distribution JS is computed before dense arrays are discarded; the compact raw artifact retains scalar metrics and distribution/state hashes.

For causal transfer, common update directions and equivalent donors are fitted only on realizations 0–1 with one clean demonstration. They are evaluated on identity-disjoint realizations 2–3 with two demonstrations and four nuisance units. Across three depths and both components, interventions include zeroing, equivalent replacement, same-semantic/different-syntax replacement, same-syntax/different-semantic replacement, fully nonequivalent replacement, common-direction projection removal, and common-direction addition (1,344 rows). This is sublayer-to-output causal mediation; it does not isolate individual attention paths.

Functional transfer appears only after training. Equivalent-minus-fully-nonequivalent target-log-probability contrasts are 0.089 nats at Pythia step 0 (family bootstrap interval crosses zero), 1.047 at step 1,000 (95% CI [0.545, 1.455]), 1.206 at step 143,000 ([0.224, 2.187]), and 0.599 for Qwen ([0.299, 0.899]). Qwen also separates equivalent from syntax mismatch by 0.574 ([0.325, 0.866]); its semantic-only mismatch contrast is 0.327 with an interval slightly crossing zero. Common-direction removal is more damaging than equivalent transfer in trained step-1,000 Pythia, final Pythia, and Qwen, supporting a task-relevant common component without claiming it is the complete mechanism. Addition is not reliably beneficial.

The variance hypothesis transfers only conditionally. Last-minus-first JS dispersion is positive at random Pythia and step 1,000, small and negative at final Pythia (-0.00135 bits, CI $[-0.00268, -0.00002]$), and positive for Qwen (0.00238, [0.00056, 0.00420]). Thus causal equivalence generalizes better than monotone nuisance denoising. The crossed fixed-factor regression is descriptive because four families are insufficient for a stable mixed-effects interaction estimate; family-clustered bootstrap is primary.

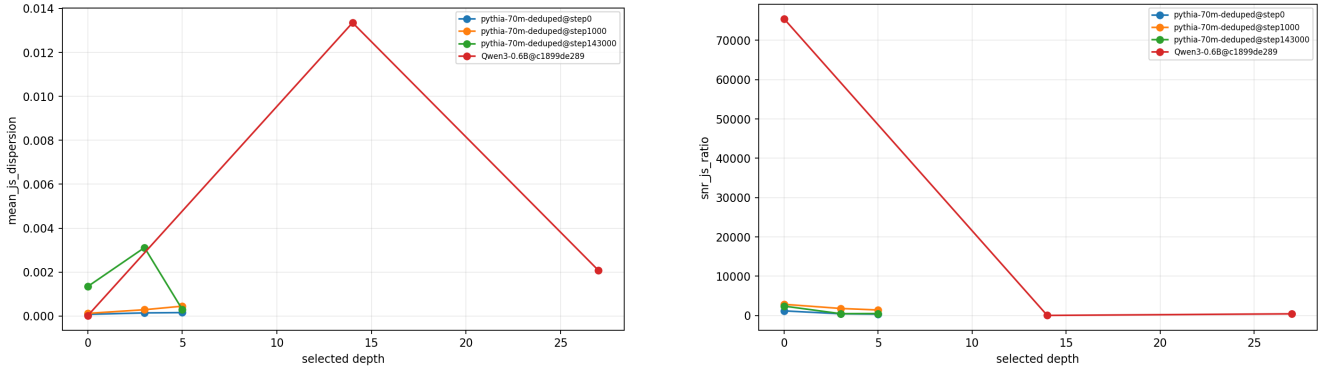


Figure 6: Pretrained fixed-depth replication under two demonstrations and four nuisance units. JS refinement is non-monotone and model/checkpoint dependent; SNR ratios are shown descriptively because near-zero denominators can dominate.

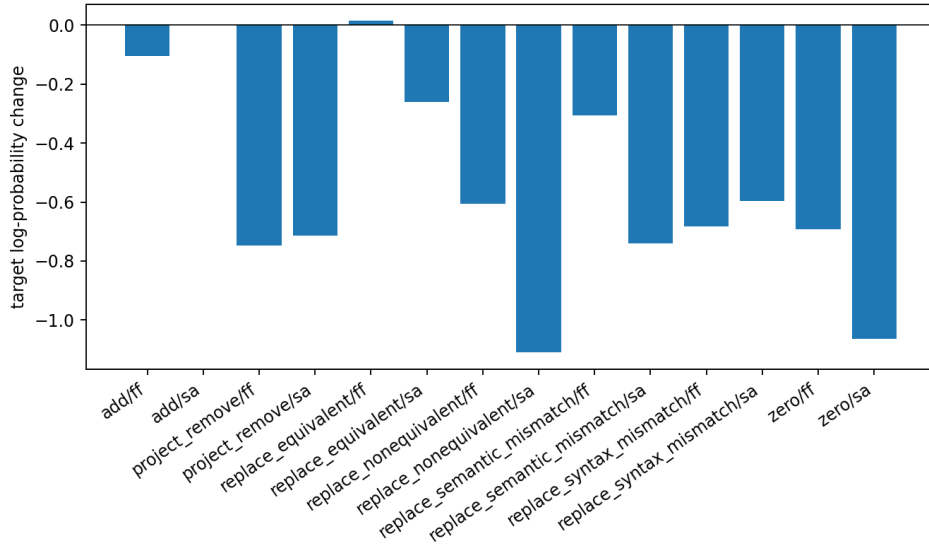


Figure 7: Held-out sublayer intervention effects across pretrained cells. Equivalent donors are fitted in a different prefix/noise regime; mismatch, zero, projection-removal, and addition controls distinguish transferable family computation from generic component utility.

13.4 Descriptive progress disagrees with intervention magnitude

Diagnostic logit progress is frequently much larger than intact-minus-ablated target effects and can rank components differently. Figure 8 retains this discrepancy instead of collapsing the measurements. This is direct evidence for the paper’s guardrail: an update aligned with the target unembedding is not thereby the causal site of the relation.

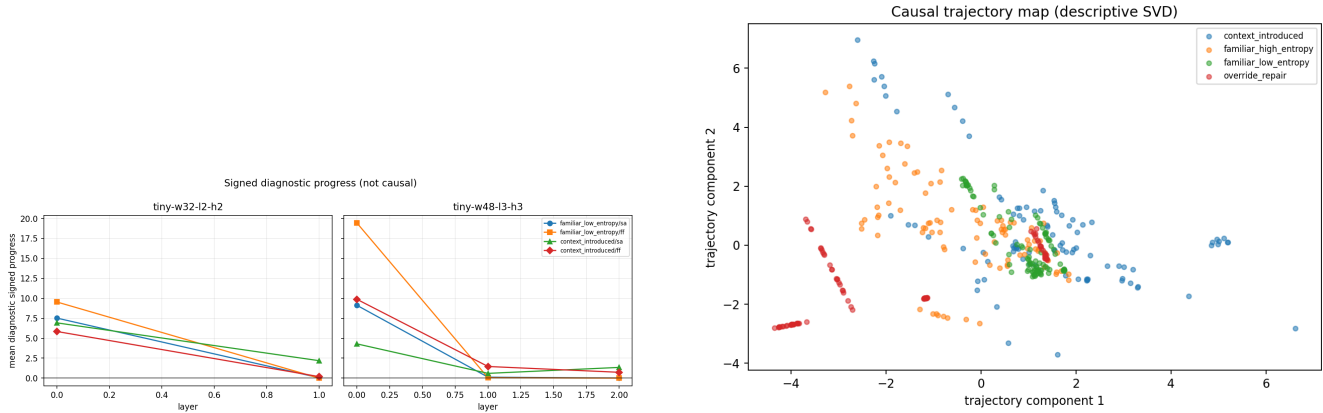


Figure 8: Left: signed diagnostic progress, explicitly not a causal effect. Right: descriptive SVD of per-example causal trajectories; cluster appearance is exploratory and no discrete mechanism classes are asserted.

13.5 Stable motifs fail the causal-utility test

Mean within-minus-control motif cosine is 0.228, so aligned attention rows are measurably relation-specific in this corpus. However, the Spearman correlation between motif specificity and SA zero-ablation contribution is -0.009 over 18 relation–layer units. E3 therefore yields a useful negative result: stability survives the matched control, but does not predict causal utility at the tested scale.

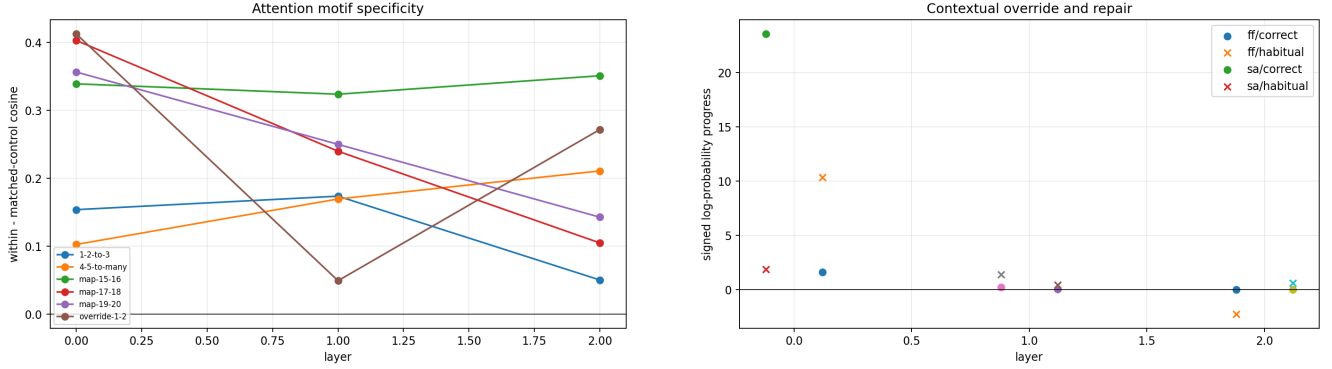


Figure 9: Left: motif specificity by relation and layer. Right: signed progress for habitual and contextually correct continuations. Geometry labels in the raw artifact remain candidates until paired with task-conditioned causal and temporal evidence.

13.6 Training dynamics and decision gate

Checkpoint measurements are non-monotonic across strata and components; the controlled run does not establish a general diffuse-to-FFN compilation trajectory. The exploratory regression of causal contribution on controlled frequency, entropy, context status, and component explains only $R^2 = 0.073$. Combined with the motif null, this fails the gate for E7 and motivates pretrained checkpoint replication rather than online adaptation.

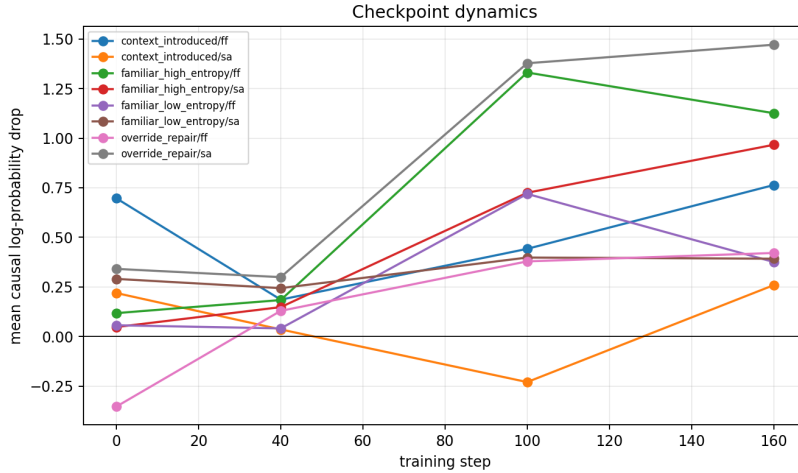


Figure 10: Component causal contribution over local training checkpoints. The trajectories are retained even when they do not match a monotone compilation account.

14 Series-Level Interpretation

Paper 0.5 establishes the first recurring pattern in the research sequence: descriptive regularity is easier to find than reusable causal structure. Relation-specific attention rows exist, yet their specificity is unrelated to SA intervention magnitude; signed logit progress can also disagree with intact-minus-ablated effects. Paper 0.6 repeats this separation in a semantic setting: hierarchy-neighbor recovery is 0.477 versus 0.254 under permutation, while semantic motif specificity is -0.012 . Paper 1 then shows what a stronger bridge must look like. Explicit cross-community bridge scores improve a controlled matched-budget selection frontier, but retained frozen-Qwen summaries add essentially no identity-disjoint predictive value after surface controls.

Together these results revise the learning hypothesis. The candidate persistent object should not be an attention motif simply because it recurs. Pretrained causal transfer now supports family-conditioned computation, but addition does not improve outcomes and Paper 1’s natural utility gate remains negative. A persistent target must

still establish matched-budget future utility and preserve contextual override. E7 therefore remains correctly stopped despite the stronger mechanistic evidence.

15 Reproducibility and Artifact Contract

The commands are

```
PYTHONPATH=src python -m cl.experiments.paper05_ngram
PYTHONPATH=src python -m cl.experiments.paper05_pretrained
PYTHONPATH=src python -m cl.analysis.paper05_inference
```

Manifests record model/checkpoint identities, hashes, direction-fit separation, and resolved configuration. Per-realization variance artifacts are split by stage/context/target control to keep files reviewable; dense distributions and states are represented by hashes after exact aggregation. Eighteen figures (15 controlled and three pretrained) and all tables regenerate from committed artifacts. Tests cover entropy/equivalence metrics, trace parity, residual identities, intervention locality, and artifact invariants.

16 Falsification Criteria

The paper should explicitly accept the following outcomes:

1. Stable attention motifs do not generalize across contexts once lexical/position controls are matched.
2. Frequency does not predict greater MLP localization or reduced SA dependence.
3. Apparent logit-lens contributions disappear under causal intervention.
4. N-gram computation is predominantly distributed and cannot be usefully categorized into trajectory families.
5. Training checkpoints show no directional migration from contextual computation toward compiled mappings.
6. SA statistics fail to provide predictive information beyond ordinary hidden-state or token-frequency statistics for the online-learning bridge.

These outcomes would constrain, rather than invalidate, the broader continuous-learning program.

17 Contribution and Limitations

The present contribution is a tested mechanistic measurement framework, controlled atlas, conditional-variance factorial, checkpoint replication, and sublayer causal-transfer result. It does not show that Transformers implement literal symbolic n-gram tables, that corpus frequency predicts compilation generally, or that functional equivalence should be consolidated online. The controlled study has two architectures/two seeds; the pretrained crossed study has four families and one run per checkpoint. Family bootstrap is therefore primary and the fixed-factor regression descriptive. Prompts and single-token targets are narrow; selected-depth hooks do not isolate individual attention paths; replacement/addition can be off-manifold; and Qwen contradicts universal depthwise variance reduction. The strongest supported statement is narrow: trained models contain transferable family-conditioned updates, while nuisance refinement is task/model dependent and stable motifs alone remain insufficient.

18 Relationship to the Research Program

Residual-stream line. Paper 0.5 supplies controlled cases for refinement, complementarity, interference, inhibition, and repair. The n-gram trajectory can be reused as a diagnostic benchmark in the broader residual paper without making that paper depend on all experiments here.

Continuous SA learning line. This paper is the empirical precursor. The later controlled bridge result shows that task-conditioned aggregate structure can be useful. A subsequent frozen-Qwen HotpotQA/QASPER intervention finds positive bridge-minus-base means at all six native-K/V budgets, but every paired interval includes zero, S5 mostly degrades, and removal-utility prediction retains negative held-out R^2 . Prototype learning and low-rank adaptation therefore remain blocked; any future learner comparison must wait for a larger fixed-protocol replication while preserving context-sensitive override.

PRA and retrieval. The connection is secondary but useful: attention motifs and graph structure may eventually inform query-facet discovery or retrieval selection. This paper should avoid depending on PRA so that the mechanistic result remains architecture-general.

19 Paper Boundary

To keep Paper 0.5 publishable and falsifiable, the mandatory scope ends after E1–E6. A working online adapter in E7 is a bonus or appendix. Full continual-learning algorithms, catastrophic-forgetting studies, deployment policies, replay/memory management, and long-horizon adaptation belong to subsequent papers.

20 Conclusion

Common token relations provide a bridge between externally measurable prediction families and internal Transformer computation. Controlled depth reduces nuisance-conditioned dispersion without requiring lower within-example entropy. In trained Pythia and Qwen, clean weak-context directions transfer causally to noisy stronger contexts: equivalent donors are much less harmful than syntax/semantic mismatches, and removing the common direction damages target likelihood. Random initialization lacks this separation. Yet pretrained dispersion trajectories are not universally denoising, addition is not beneficial, motifs do not predict causal utility, and the downstream natural learning gate remains closed. Functional equivalence is therefore a validated mechanistic object, not yet a persistent-learning target.

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